

PRESIDENCY UNIVERSITY

CSE208: Digital Logic Design Laboratory

Lecture 07: Demultiplexer (DEMUX)

Data Distribution • Selection Lines • Boolean Design
Logic ICs • Circuit Construction • Laboratory Verification

Afzal Hossan
Lecturer
Department of Computer Science and Engineering
Presidency University

What you will learn today

LAB 7 | COURSE OUTCOME CO3

Understand a DEMUX

Explain what the data input, select lines, enable and outputs do.

Build a working circuit

Use gates to build a 1-to-4 DEMUX and an IC for 1-to-8.

Design the logic

Read truth tables, derive equations, and combine minterms to implement functions.

Prove that it works

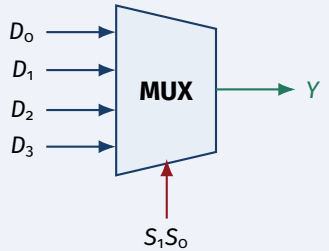
Test each selection and explain any incorrect result.

Key idea Two key skills: distribute a signal and implement Boolean functions using DEMUX outputs.

From multiplexer to demultiplexer

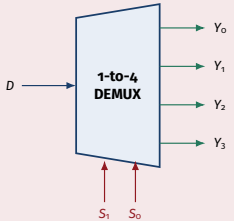
THE SELECTION DIRECTION IS REVERSED

Lecture 6: Multiplexer (MUX)



Many inputs ⟶ one output

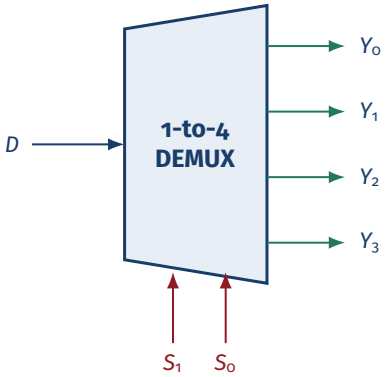
Lecture 7: Demulticplexer (DEMUX)



One input ⟶ one of many outputs

Key idea The select lines determine *which path* is used. Neither circuit needs a clock or stores data.

Identify the signals on a DEMUX



D: **data input** — the 0 or 1 being distributed.

S_1, S_0: **select lines** — choose the destination.

Y_0–Y_3: **outputs** — only the chosen line can follow *D*.

Rule: n selection bits choose among 2^n output lines.

DEMUX	1-to-2	1-to-4	1-to-8	1-to-16
Select bits	1	2	3	4

What do the select lines, data and enable do?

Select chooses *where*. **Data** determines *what*. **Enable** allows *whether*.

Ideal active-HIGH enable E

E	Effect
0	All outputs LOW
1	Selected output follows D

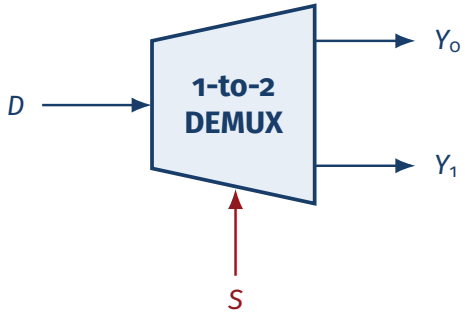
Active-LOW enable $\overline{G}$

An active-LOW enable is ON when its input is **0**, not 1.

The real IC's output levels must be checked in its datasheet.

Example: If $D = 0$, the selected output of an ideal active-HIGH DEMUX is still 0. Selecting a line does not automatically make it HIGH.

1-to-2 DEMUX: the basic idea



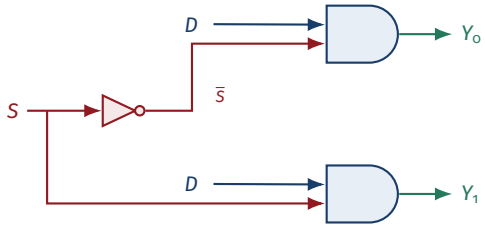
D	S	Y_0	Y_1
0	0	0	0
0	1	0	0
1	0	1	0
1	1	0	1

$$\boxed{Y_0 = D\overline{S}} \qquad \boxed{Y_1 = DS}$$

$S = 0$: send D to Y_0. $S = 1$: send D to Y_1.

Build a 1-to-2 DEMUX with logic gates

ONE NOT GATE AND TWO AND GATES



Upper AND gate

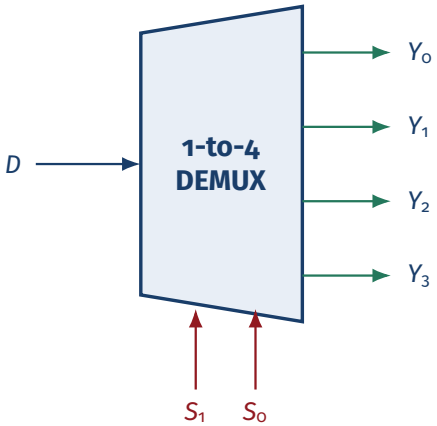
$D\bar{S}$ produces Y_0.

Lower AND gate

DS produces Y_1.

The two wires labeled D connect to the *same* data input. No OR gate is required.

1-to-4 DEMUX: two selection bits



S_1	S_0	Selected output
0	0	Y_0
0	1	Y_1
1	0	Y_2
1	1	Y_3

Example: $S_1S_0 = 10_2 = 2$. This selects Y_2 (whose value is D).

1-to-4 DEMUX: complete truth table

ALL EIGHT INPUTS ARE TESTED | ACTIVE-HIGH OUTPUTS, ENABLED

S_1	S_0	D	Y_0	Y_1	Y_2	Y_3
0	0	0	0	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	0	0
0	1	1	0	1	0	0
1	0	0	0	0	0	0
1	0	1	0	0	1	0
1	1	0	0	0	0	0
1	1	1	0	0	0	1

When $D = 1$
Exactly one output is HIGH.

When $D = 0$
All outputs are LOW.

Key idea If the active-HIGH enable is OFF, every output is LOW regardless of D, S_1, S_0.

Where do the 1-to-4 equations come from?

AN OUTPUT IS HIGH ONLY WHEN ITS ADDRESS MATCHES AND $D = 1$

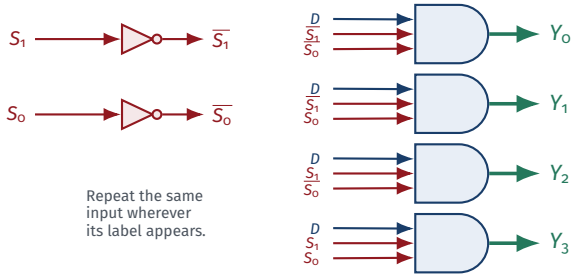
Select S_1S_0	Condition
00	$\overline{S_1S_0}$
01	$\overline{S_1}S_0$
10	$S_1\overline{S_0}$
11	S_1S_0

$$Y_0 = D\overline{S_1S_0}$$
$$Y_1 = D\overline{S_1}S_0$$
$$Y_2 = DS_1\overline{S_0}$$
$$Y_3 = DS_1S_0$$

Example: Output 2 needs select 10, so $S_1 = 1$, $S_0 = 0$. Thus $Y_2 = DS_1\overline{S_0}$.

Gate-level circuit for a 1-to-4 DEMUX

TWO NOT GATES AND FOUR 3-INPUT AND FUNCTIONS



Each AND gate receives *three* signals: *D* plus the two select terms for its output.

Build the gate circuit with real ICs

74HC04 INVERTER + TWO 74HC08 AND CHIPS

Stage 1: generate four matches

$M_0 = \overline{S_1} S_0$

$M_1 = \overline{S_1} S_0$

$M_2 = S_1 \overline{S_0}$

$M_3 = S_1 S_0$

Stage 2: pass the data

$Y_0 = DM_0$

$Y_1 = DM_1$

$Y_2 = DM_2$

$Y_3 = DM_3$

Total: one 74HC04 (use 2 NOT gates) + two 74HC08 chips (use 8 two-input AND gates). No OR gate is needed.

Worked example: follow the selected output

PREDICT, THEN VERIFY

Given $D = 1$, $S_1S_0 = 10$.

Step 1: $10_2 = 2$, so select Y_2.

Step 2: $Y_2 = DS_1\overline{S_0}$

$= 1 \times 1 \times 1 = 1.$

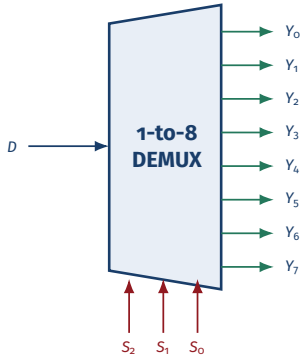
Result

$$\boxed{(Y_0, Y_1, Y_2, Y_3) = (0, 0, 1, 0)}$$

Change only D to 0.

The selection is still output 2, but *every* output now reads 0.

1-to-8 DEMUX: three selection bits



Select	Active output	Select	Active output
000	$Y_0 = D$	100	$Y_4 = D$
001	$Y_1 = D$	101	$Y_5 = D$
010	$Y_2 = D$	110	$Y_6 = D$
011	$Y_3 = D$	111	$Y_7 = D$

S_2 is the most significant bit. S_0 is the least significant bit.

Derive any 1-to-8 output equation

READ THE BINARY OUTPUT NUMBER

Example: derive Y_5. $5 = 101_2$.

That address requires $S_2 = 1$, $S_1 = 0$, $S_0 = 1$.

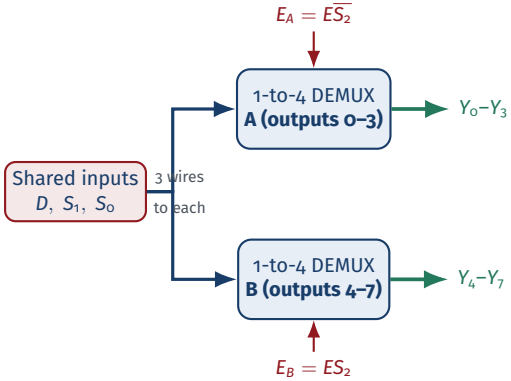
$$\boxed{Y_5 = ED\,S_2\overline{S_1}S_0}$$

Repeat the rule for any Y_k:

1. Convert k to a 3-bit binary address.
2. Complement the select bits that must be 0.
3. AND the select terms with D (and E if used).

Build 1-to-8 using two 1-to-4 DEMUXs

ONE EXTRA SELECT BIT CHOOSES ONE OF TWO OUTPUT GROUPS



$S_2 = 0$: A ON
A controls Y_0–Y_3.

$S_2 = 1$: B ON
B controls Y_4–Y_7.

Both use the *same* D, S_1, S_0. If $E = 0$, both are OFF.

Example: which unit and which output?

INPUTS: $E = 1$, $D = 1$, $S_2S_1S_0 = 101$

Step 1: read $S_2 = 1$.
Unit B is ON; unit A is OFF.

Step 2: read $S_1S_0 = 01$.
Choose local output 1 in unit B.

Step 3: map to final output.
Unit B starts at Y_4, so $Y_{4+1} = Y_5$.

Final result

$$\boxed{Y_5 = 1}$$

All other outputs = 0.

Check:
If $D = 0$, all outputs become 0, although unit B remains selected.

New application: a DEMUX can create Boolean functions

FIX THE DATA INPUT AT 1 TO GENERATE MINTERMS

Recall the 1-to-4 equations

$Y_0 = D\overline{S_1}\overline{S_0}$

$Y_1 = D\overline{S_1}S_0$

$Y_2 = DS_1\overline{S_0}$

$Y_3 = DS_1S_0$

Now set $D = 1$

$Y_0 = \overline{A}\overline{B}$

$Y_1 = \overline{A}B$

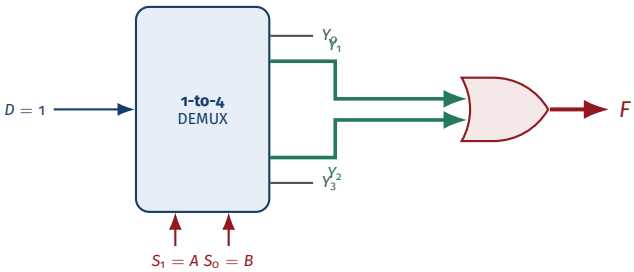
$Y_2 = A\overline{B}$

$Y_3 = AB$

Make the connection: use $S_1 = A$ and $S_0 = B$. Each output is now one *minterm* (an AND term for exactly one input combination). An OR gate can combine the minterms needed by a function.

Example 1: implement XOR using a 1-to-4 DEMUX

SELECTED OUTPUTS Y_1 AND Y_2 ARE CONNECTED TO AN OR GATE



Target: XOR

$$F = A \oplus B$$
$$= \bar{A}B + A\bar{B}$$
$$= \boxed{Y_1 + Y_2}$$

Check: $AB = 01$ activates Y_1; $AB = 10$ activates Y_2.

XOR is HIGH only in those two cases.

Many Boolean functions from the same 1-to-4 DEMUX

KEEP $D = 1$, $S_1 = A$, $S_0 = B$; CHANGE ONLY THE OUTPUT CONNECTIONS

Function	Boolean expression	Combine DEMUX outputs
AND	AB	Y_3
OR	$A + B$	$Y_1 + Y_2 + Y_3$
XOR	$A \oplus B$	$Y_1 + Y_2$
XNOR	$\overline{A \oplus B}$	$Y_0 + Y_3$
NOT A	$\overline{A}$	$Y_0 + Y_1$

General rule: a 1-to-2^n DEMUX with $D = 1$, plus suitable OR gates, can implement any Boolean function of its n select variables.

Assumption: these equations use *active-HIGH* outputs. Active-LOW hardware needs a different combining gate (shown later).

Example 2: implement a three-variable function

USE A 1-TO-8 DEMUX AND SELECT THE REQUIRED MINTERMS

Target function

$F(A, B, C) = \Sigma m(1, 2, 6, 7)$

Meaning: $F = 1$ for input codes 001, 010, 110, 111.

ABC	000	001	010	011	100	101	110	111
F	0	1	1	0	0	0	1	1

Connect the DEMUX

$D = 1$

$S_2 = A,\ S_1 = B,\ S_0 = C$

OR these outputs:

$$\boxed{F = Y_1 + Y_2 + Y_6 + Y_7}$$

Example: $ABC = 110$ selects $Y_6 = 1$; therefore $F = 1$.

$ABC = 101$ selects Y_5, which is not ORed; $F = 0$.

A 4-input OR function can be built from three 2-input OR gates (two ORs, then a final OR).

Example 3: full-adder truth table gives two functions

CONNECT TO THE FULL-ADDER CIRCUIT FROM LAB 4

A	B	C_{in}	Sum	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Sum is HIGH for

001, 010, 100, 111

$\text{Sum} = \Sigma m(1, 2, 4, 7)$

Carry is HIGH for

011, 101, 110, 111

$C_{out} = \Sigma m(3, 5, 6, 7)$

Question: Can we produce *both* outputs with one 1-to-8 DEMUX? Yes: create all eight minterms once and combine different groups.

One 1-to-8 DEMUX can implement a full adder

THE SAME MINTERM OUTPUTS ARE SHARED BY TWO SEPARATE OR NETWORKS

Common connections: $D = 1,\ S_2 = A,\ S_1 = B,\ S_0 = C_{in}$.

OR network 1: SUM

$Y_1 \quad Y_2 \quad Y_4 \quad Y_7$

$$\boxed{\text{Sum} = Y_1 + Y_2 + Y_4 + Y_7}$$

OR network 2: CARRY

$Y_3 \quad Y_5 \quad Y_6 \quad Y_7$

$$\boxed{C_{out} = Y_3 + Y_5 + Y_6 + Y_7}$$

Trace one case: $ABC_{in} = 110$ activates Y_6. Y_6 is used by Carry but not Sum: $(\text{Sum}, C_{out}) = (0, 1)$.

Y_7 feeds *both* OR networks; branching one output into two logic-gate inputs is allowed within IC fan-out limits. Never tie two logic *outputs* directly together.

Real ICs: check the output polarity

WE WILL USE ONE OF THESE 16-PIN ICS

CD74HC238: active-HIGH

Selected output becomes **1** when $D = 1$.

Other outputs remain 0.

Best for the ideal truth table.

SN74HC138: active-LOW

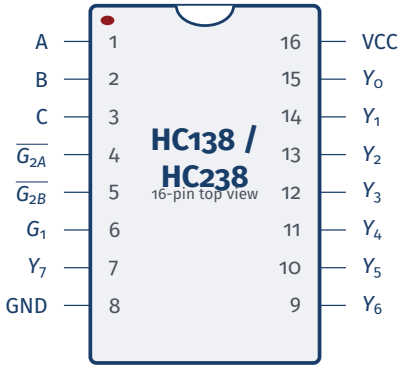
Selected output becomes **0** when $D = 1$.

Other outputs remain 1.

Same routing, opposite electrical level.

Both devices: address $A = S_0$, $B = S_1$, $C = S_2$. Enable with pin 6 HIGH and pins 4, 5 LOW. For the DEMUX experiment, let pin 6 carry D.

IC pin diagram: see the actual connections



Power
V_{CC} = pin 16 GND = pin 8

Select
S_0 = pin 1 S_1 = pin 2 S_2 = pin 3

Enable/data
Pins 4 and 5 ← 0
Pin 6 ← D

Output pins: $Y_0 = 15$, $Y_1 = 14$, $Y_2 = 13$, $Y_3 = 12$, $Y_4 = 11$, $Y_5 = 10$, $Y_6 = 9$, $Y_7 = 7$.
Top view, notch upward. Check the exact package datasheet before wiring.

Wire the CD74HC238 as a 1-to-8 DEMUX

FOLLOW THIS CONNECTION TABLE WITH POWER OFF

Pins	Connect to
16	Regulated +5 V
8	Common GND
1, 2, 3	S_0, S_1, S_2
4, 5	Logic 0 (LOW)
6	Data input D
15,14,13,12	Indicators Y_0–Y_3
11,10,9,7	Indicators Y_4–Y_7

Try it

$D = 1$ and $S_2S_1S_0 = 101$

Output Y_5 (pin 10) goes HIGH.

Next

Keep select = 101; change D to 0. All outputs go LOW.

HC238 LED wiring: $Y_k \rightarrow 1\,\mathrm{k}\Omega \rightarrow \mathrm{LED} \rightarrow \mathrm{GND}$. LED anode faces the output. A selected HIGH lights the LED.

If the lab has a 74HC138 instead

THE SELECTED OUTPUT GOES LOW, NOT HIGH

When $D = 1$

The selected output is 0; the other seven outputs are 1.

When $D = 0$

All eight outputs are 1 (device disabled).

LED indicator connection

$+5\,\mathrm{V} \rightarrow R \rightarrow LED \rightarrow Y_k$

A LOW output *sinks* current, so the selected LED turns ON.

Example: $D = 1$ and select 101 makes Y_5 (pin 10) LOW on HC138, whereas HC238 makes the same selected pin HIGH.

Use a resistor (for example 1 kΩ at 5 V) with each low-current LED. Check the output-current rating of the particular IC.

Real-IC detail: OR for HC238, NAND for HC138

THE SAME FUNCTION, BUT THE OUTPUT POLARITIES DIFFER

CD74HC238: active-HIGH

Selected minterm output $Y_k = 1$.

For $F = \Sigma m(1, 2, 6, 7)$:

$F = Y_1 + Y_2 + Y_6 + Y_7$

Combine with an OR network.

SN74HC138: active-LOW

Selected output $L_k = 0$; others $= 1$.

For the same function:

$F = \overline{L_1 L_2 L_6 L_7}$

Combine with a 4-input NAND.

Why NAND works: if any required minterm is selected, its L_k becomes 0, so NAND gives 1. For any other selection all four listed $L_k = 1$, so NAND gives 0.

Example ICs: SN74HC32 (four 2-input OR gates) and SN74HC20 (two 4-input NAND gates). Use a common compatible supply and verify the datasheets before construction.

Gate ICs used in the 1-to-4 experiment

BOTH 14-PIN IC TYPES USE PIN 14 = +5 V, PIN 7 = GROUND

SN74HC04: six NOT gates

Input	Inverted output
1	2
3	4
5	6
9	8
11	10
13	12

SN74HC08: four 2-input ANDs

Inputs	AND output
1, 2	3
4, 5	6
9, 10	8
12, 13	11

Key idea We need **one** HC04 and **two** HC08 ICs to build the 1-to-4 DEMUX with ordinary 2-input gates.

1-to-4: connect the gates in three steps

U0 = HC04; U1 AND U2 = HC08

1. U0: make the complements

$S_1 \rightarrow$ pin 1; $\overline{S_1}$ from pin 2.

$S_0 \rightarrow$ pin 3; $\overline{S_0}$ from pin 4.

2. U1: make the select matches

(1,2)$\rightarrow$3: $M_0 = \overline{S}_1\overline{S}_0$
(4,5)$\rightarrow$6: $M_1 = \overline{S}_1 S_0$
(9,10)$\rightarrow$8: $M_2 = S_1\overline{S}_0$
(12,13)$\rightarrow$11: $M_3 = S_1 S_0$

3. U2: distribute *D*

(1,2)$\rightarrow$3: $Y_0 = DM_0$
(4,5)$\rightarrow$6: $Y_1 = DM_1$
(9,10)$\rightarrow$8: $Y_2 = DM_2$
(12,13)$\rightarrow$11: $Y_3 = DM_3$

Supply each IC: pin 14 = +5 V; pin 7 = GND. Connect *D* to one input of *each* gate on U2.

Before switching on: five safety checks

1. Turn **power OFF** before changing wires.
2. Match notch, pin 1 and the **actual IC** marking.
3. Connect the correct **VCC and ground** pins.
4. Give every input a definite 0 or 1.
5. Use **current-limiting resistors** for LEDs.

Never

Short an output directly to +5 V or ground.

Apply power while moving an IC.

Recommended

Place a 0.1-μF bypass capacitor near each IC's VCC/GND pins.

Experiment 1: verify the gate-built 1-to-4

EXPECTED RESULT: THE TRUTH TABLE MUST MATCH FOR ALL 8 CASES

Procedure

1. Build U0, U1 and U2 using the connection map.
2. Connect data/select switches and output indicators.
3. Apply $S_1S_0 = 00, 01, 10, 11$ at $D = 1$.
4. Repeat all four selections with $D = 0$.
5. Record expected and measured outputs.

D	S_1S_0	Expected HIGH output
1	00	Y_0
1	01	Y_1
1	10	Y_2
1	11	Y_3
0	any	None

Extra check: with select = 10, toggle only D. Only Y_2 should follow it.

Experiment 2: verify the IC-based 1-to-8

USE CD74HC238 AS THE PRIMARY ACTIVE-HIGH IMPLEMENTATION

Procedure

1. Wire the IC using the pin-connection slide.
2. Fix $D = 1$; test all eight addresses.
3. Fix $D = 0$; repeat all eight addresses.
4. Disable the IC by taking pin 4 HIGH.
5. Record what happens to each output.

Select	HIGH	Select	HIGH
000	Y_0	100	Y_4
001	Y_1	101	Y_5
010	Y_2	110	Y_6
011	Y_3	111	Y_7

Expected: $D = 0$ or disabled $\Rightarrow$ all HC238 outputs LOW.

Optional experiment 3: build XOR with a DEMUX IC

CD74HC238 + ONE OR GATE FROM SN74HC32

CD74HC238 inputs

Pin 16 $\rightarrow$ +5 V, pin 8 $\rightarrow$ GND.
Pin 3 (S_2) $\rightarrow$ 0.
Pin 2 (S_1) $\rightarrow$ A.
Pin 1 (S_0) $\rightarrow$ B.
Pins 4,5 $\rightarrow$ 0; pin 6 (D) $\rightarrow$ 1.

SN74HC32 OR gate

HC238 pin 14 (Y_1) $\rightarrow$ HC32 pin 1.
HC238 pin 13 (Y_2) $\rightarrow$ HC32 pin 2.
Read $F = A \oplus B$ on HC32 pin 3.
HC32 pin 14 $\rightarrow$ +5 V, pin 7 $\rightarrow$ GND.

Expected truth table

A	B	$F = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Lab rules: use one common ground, never leave CMOS inputs floating, connect an LED through a resistor, and switch power off before rewiring.

What students must record

DO NOT REPORT ONLY “THE CIRCUIT WORKED”

1-to-4 observation sheet

D	S_1S_0	Expected	Measured
1	00	1000	_____
1	01	0100	_____
1	10	0010	_____
1	11	0001	_____
0	00–11	0000	_____

Vector order: $Y_0Y_1Y_2Y_3$.

1-to-8 observation sheet

D	Address	Active line	Measured
1	000	Y_0	_____
1	001	Y_1	_____
⋮	⋮	⋮	⋮
1	111	Y_7	_____
0	all 8	none	_____

Include a separate disabled-enable test.

Key idea Expand the grouped rows into individual tests in the laboratory notebook.

Where is a DEMUX useful?

THREE PRACTICAL EXAMPLES

LED / device selection

Select one of several control lines. *D* sets its active level.

Memory-chip enable

Address bits choose a device to enable. The multibit data bus is separate.

Boolean functions

OR selected minterms to implement XOR or full-adder logic.

Limit: A DEMUX distributes a logic signal; it does not automatically store data or transmit an entire multi-bit bus.

Troubleshooting: find the error systematically

What you see	Possible cause	First check
No output changes	Power or enable incorrect	VCC, GND, pins 4–6
Wrong line is active	Select inputs exchanged	S_0, S_1, S_2 order
All outputs HIGH (HC138)	Device disabled / $D = 0$	Pin 6 and enables
All outputs LOW (HC238)	Device disabled / $D = 0$	Pin 6 and enables
Unstable LED / output	Floating or bouncing input	Logic switches / pull resistors
Wrong gate-built output	Miswired NOT or AND gate	$\overline{S}_1, \overline{S}_0, M_i$

When a test fails: write down the input state, actual outputs and cause, then correct the connection and retest.

What to submit in the laboratory report

Design evidence

1. Objective and component list.
2. Labeled circuit diagram.
3. Truth table and Boolean equations.
4. Actual IC pin connections.

Experimental evidence

5. Expected *and* measured outputs.
6. Explanation of active output polarity.
7. Any error and its correction.
8. Short conclusion supported by tests.

Key idea The result is verified only when the observed values agree with the circuit's correct truth table.

Quick viva and final summary

Answer aloud

1. Why does 1-to-8 require three selects?
2. With $D = 1$ and $S_1S_0 = 11$, which 1-to-4 output is HIGH?
3. Derive Y_2 of a 1-to-4 DEMUX.
4. Why is the selected HC138 output LOW?
5. How does $Y_1 + Y_2$ implement XOR?

Remember

- DEMUX: one data input, many outputs.
- Select lines choose a destination.
- The selected output follows D.
- With $D = 1$, DEMUX outputs become minterms.
- Real ICs may use active-LOW signals.

Exit task: Draw a 1-to-4 DEMUX, mark its inputs and outputs, and explain one truth-table row.

References and official datasheets

Presidency University, Department of CSE. *CSE208: Digital Logic Design Laboratory Course Outline*. Lab 7: Demultiplexer; CO3.

Mano, M. M. and Ciletti, M. D. (2018). *Digital Design*, 6th ed. Pearson.
Tocci, R. J., Widmer, N. S. and Moss, G. L. (2014). *Digital Systems: Principles and Applications*, 11th ed. Pearson.

Texas Instruments: Product datasheets and function tables.
https://www.ti.com/product/CD74HC238
https://www.ti.com/product/SN74HC138
https://www.ti.com/product/SN74HC04 https://www.ti.com/product/SN74HC08
https://www.ti.com/product/SN74HC32 https://www.ti.com/product/SN74HC20